Maddi Rishi Dhaneshwar

Email: maddi.rishi2468@gmail.com Phone: +91 9704670400
GitHub: https://github.com/RishiMaddi1 LinkedIn: www.linkedin.com/in/rishi-maddi

Education

B.Tech in Artificial Intelligence and Data Science — Amrita University, Bangalore 2023 - 2027

Technical Skills

Programming Languages: Python, Java, JavaScript, Dart, C, PostgreSQL, PL/SQL
Web Frameworks: Flutter, Node.js, Express.js, Flask, Fast API, Spring Boot
AI/ML Technologies: Transformers, Large Language Models, Reinforcement Learning, Deep Learning
Database Systems: PostgreSQL, Firebase, MongoDB, Redis
Backend Technologies: RESTful APIs, Firebase Authentication, Microservices, Caching
Development Tools: Git, Docker, Visual Studio Code, Postman, Cloudflare, Supabase, Azure DevOps

Professional Experience

SDE Intern – Bajaj Finserv Health Limited, Bengaluru India – September 2025 - Present
- Optimized PL/SQL procedures and reduced DB calls by 50% on bulk enrollment operations
- Implemented startup-time caching strategy eliminating 2.5 Lakh+ daily redundant DB lookups
- Architected a multi-queue event system achieving zero blocking during peak traffic scenarios
- Optimized document upload pipeline reducing DB calls from N per document to 1 per batch

Applied AI and Automation Intern – PavePilot AI, Delaware, USA – June 2025 - July 2025
- Built automation pipelines integrating 5 AI platforms (OpenAI, Replicate, HeyGen, ElevenLabs, Azure OpenAI) automating workflows by 40%
- Automated video editing and e-commerce extraction, cutting end-to-end processing time by 3x

Full Stack Developer Intern – PavePilot AI, Delaware, USA – August 2024 - September 2024
- Transformed 15+ Figma designs into responsive Flutter interfaces with 100% design fidelity
- Architected PostgreSQL schema and RESTful APIs
- Executed testing using Postman, resolving 25+ critical bugs pre-deployment

Full Stack Developer Intern – Thrupera – July 2024 - September 2024
- Implemented file management system handling 10GB+ transfers with 50% faster speeds and 30% improved user engagement

Achievements

YC Startup School '26 India Cohort
Selected participant; awarded $25,000 in AI credits

National Finalist — HackRX 6.0 Bajaj Finserv Health, 2025
Finalist performance led to SDE internship offer

National Finalist — AI Impact Buildathon AI Impact Summit, New Delhi, 2026

Notable Projects

Aikipedia.in (Founder & CEO)
- Launched fully operational SaaS platform with affordable AI model access, generating revenue through pay-per-use credit system
- Architected complete technology ,implementing Cloudflare hosting with Supabase backend and edge functions for razorpay payments achieving 100% uptime
- Democratized AI accessibility in India by reducing costs by 80% compared to direct API access, enabling seamless project development

Cloud-Federated Multi-GPU Distributed RAG System (Sept 2025 - Nov 2025)
- Architected a decentralized RAG pipeline that parallelizes embedding generation and vector search across a distributed mesh of consumer laptops via FastAPI and local network orchestration
- Optimized task sharding for heterogeneous hardware, enabling near-linear speedups while eliminating the need for expensive centralized server infrastructure

RL-CACHE: PostgreSQL Optimization (Apr 2025 - May 2025)

- Engineered a 3-tier intelligent caching architecture using reinforcement learning to optimize PostgreSQL query performance by 60% compared to standard configurations
- Built a real-time simulation environment to demonstrate performance superiority against baseline systems under dynamic, multi-seasonal demand patterns

Clip2Learn: Intelligent Tutorial Search with Timestamps and Visual Context (Oct 2025 - Nov 2025)
- Designed a scalable Big Data + NLP pipeline to extract step-level answers from YouTube tutorials, delivering accurate timestamped explanations
- Implemented an end-to-end flow: ingestion, Whisper transcription, vision-based frame understanding, SentenceTransformers embeddings, FAISS retrieval, and RAG answer generation with a Gradio interface

A Graph-Structured Memory Framework for Context-Aware Conversational Agents (Sept 2025 - Nov 2025)
- Architected a Graph-RAG persistent memory system using Neo4j and FastAPI, injecting high-signal context via PageRank and Shannon entropy
- Engineered a scalable two-tier retrieval pipeline with sub-second latency for 50k+ nodes, featuring a custom fallback mechanism and linear scaling to 100k+ nodes
- Optimized long-term context retention using Louvain community detection and LLM-based summarization, achieving 40-60% graph compression while preserving semantic and temporal coherence

Cognitive-Cane: IoT Navigation Assistant for the visually impaired
- Designed real-time navigation system processing visual data to generate audio instructions for visually impaired users
- Integrated sensors with cloud processing, achieving 5-second response time for interaction

Hybrid PCA-HCPSA Spectrogram Compression Framework (Accepted for Publication by the American Institute of Physics)
- Pioneered novel compression algorithm achieving 80% size reduction while maintaining 95% reconstruction quality
- Optimized performance through Mean Squared Error metrics, outperforming traditional compression methods